

individual-assignment5

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Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(janitor)
# file organization
library(here)
# wrapping long axis labels
library(scales)
# reading in .xlsx files
library(readxl)
# visualizing missing data
library(naniar)
# arranging multiple plots
library(patchwork)
# converting to snake case
library(snakecase)
```

Data

```
# taxonomic information for each taxon ID
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrate count data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```

        sheet = "Aquatic Insects")

# site metadata
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")

# water quality measurements
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
                          sheet = "Water Quality",
                          na = c("", "n/a", "N/A", "over", "173+"))

```

Class Code

```

# filter sites to only quarterly NCOS sites for downstream filtering joins
ncos_sites <- sites |>
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# convert taxon names to snake_case so they match column names in aq_ins
taxon_list_clean <- taxon_list |>
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

```

# create summarized water quality data frame for each date-site combination
water_quality_clean <- water_quality |>
  # standardize column names
  clean_names() |>
  # keep only NCOS quarterly sites
  semi_join(ncos_sites, by = "site") |>
  # create a single date_site key for joining
  unite("date_site", date, site, remove = TRUE) |>
  # group by survey event
  group_by(date_site) |>
  # calculate median water quality metrics per survey event
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # remove a known salinity outlier

```

```
filter(date_site != "2022-08-12_NVBR")
```

```
# create main analysis data frame: long format with one row per
# taxon per survey event, with water quality and taxonomy joined in
aq_ins_clean <- aq_ins |>
  # standardize column names
  clean_names() |>
  # keep only NCOS quarterly sites
  semi_join(ncos_sites, by = c("site")) |>
  # rename date column for clarity
  rename(date = date_on_vial) |>
  # select only relevant taxon and metadata columns
  select(date, sample_type, site,
         ostracod, copepod,
         hemiptera_corixidae_boatman,
         cladocera, nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # keep only filtered beaker samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # pivot to long format so each row is one taxon per survey
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by survey event and taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (total count / 7.5 L)
  summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  # ungroup data frame
  ungroup() |>
  # create date_site key for joining with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # attach water quality measurements
  left_join(water_quality_clean, by = "date_site") |>
  # attach full taxonomic information
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add human-readable site names
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
```

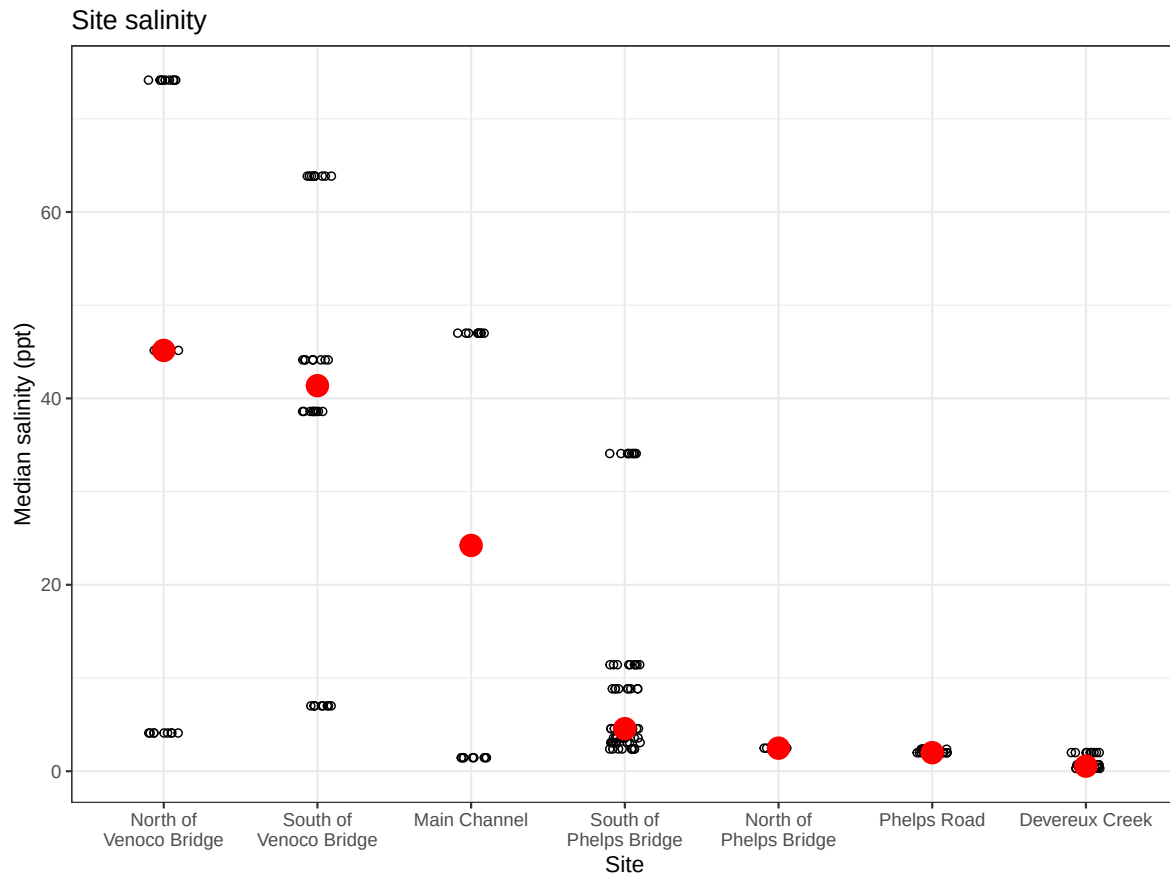
```

    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~ "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # order sites by median salinity (high to low) for consistent axis ordering
  mutate(site_full = fct_reorder(site_full, med_sal,
                                .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))

# salinity panel: one point per survey, median shown in red
salinity <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = site_full, y = med_sal)) +
  # individual survey salinity measurements
  geom_jitter(height = 0, width = 0.1, shape = 21) +
  # wrap long site names on x-axis
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # red point for median salinity per site
  stat_summary(fun = median, color = "red", size = 1) +
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  theme_bw()

salinity

```



```
# filter to copepods only and log+1 transform abundance
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  mutate(log_ave_lit = log(ave_lit + 1))
```

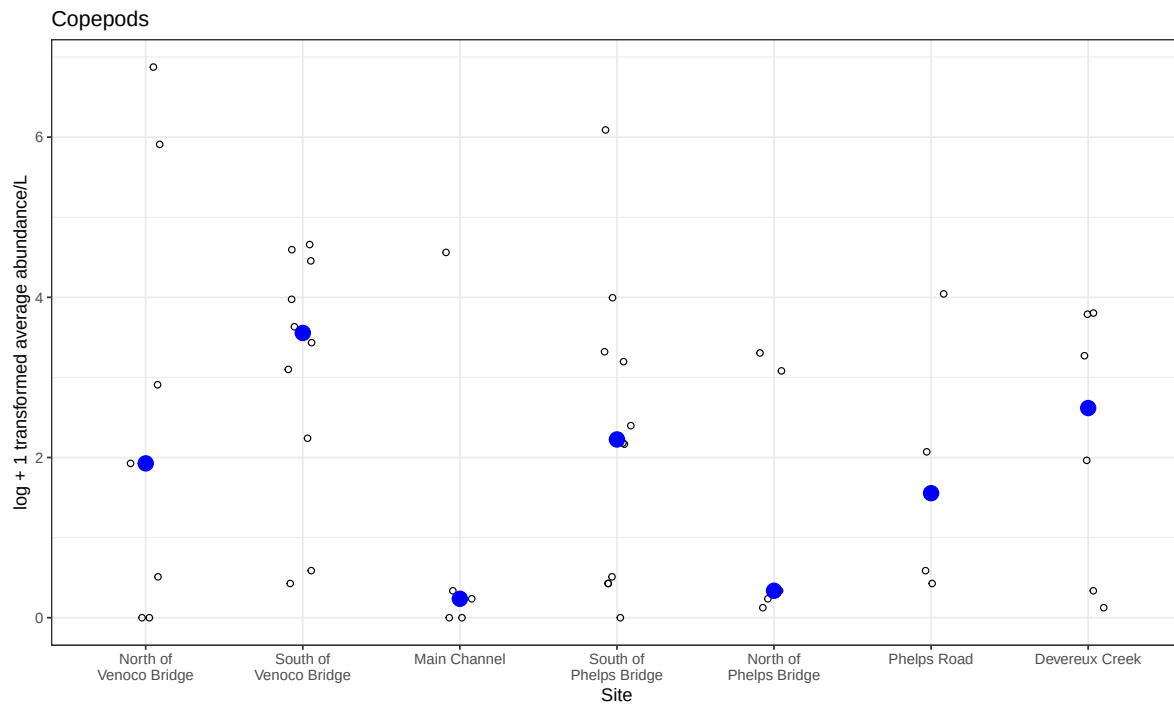
```
# copepod panel: individual observations + blue median point
copepods_plot <- ggplot(data = copepods,
  mapping = aes(x = site_full, y = log_ave_lit)) +
  # individual observations as open circles
  geom_jitter(height = 0, shape = 21, width = 0.1) +
  # blue point for median per site
  stat_summary(geom = "point", fun = "median",
    size = 4, color = "blue") +
  scale_x_discrete(labels = label_wrap(15)) +
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
```

```

    title = "Copepods") +
  theme_bw()

copepods_plot

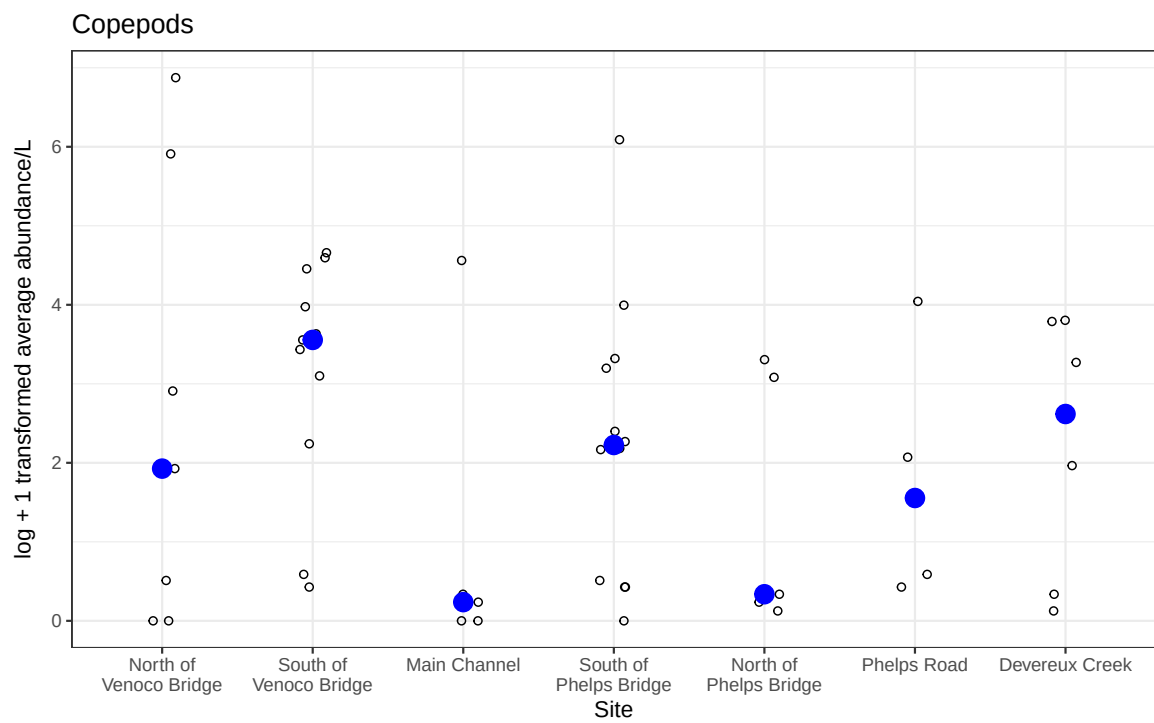
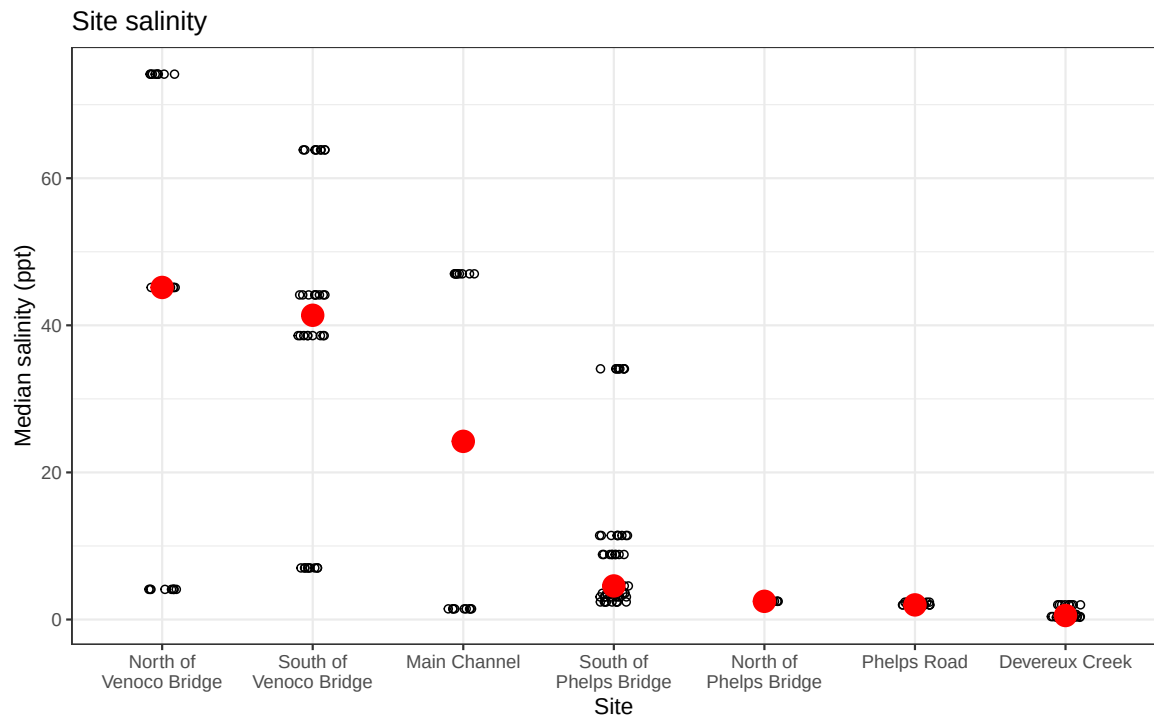
```



```

# two-panel figure from class: salinity stacked above copepods
salinity / copepods_plot

```



1. Visualizing differences across sites in major taxon abundances

a. Planning the figure

- x axis: `site_full` (site name)
- y axis: log + 1 transformed average ostracod abundance per liter
- geometry to represent average abundance/L: `geom_jitter()` to plot every individual survey observation as points
- geometry to represent medians: `stats_summary(fun = "median")` overlays colored points to show medians per site

b. Wragling Data

- filter `aq_ins_clean` to only include ostracods
- log +1 transform average abundances

```
# creating new object from aq_ins_clean
ostracods <- aq_ins_clean |>
  # filter to only include the taxon titled ostracod
  filter(taxon == "ostracod") |>
  # creates new column called log_ave_lit that contains the log + 1
  #   transformed average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2021-08-25_NPB	2021-08-25 00:00:00	NPB	ostr~	581.	7.52	4.57	4.5
2	2023-05-06_NDC	2023-05-06 00:00:00	NDC	ostr~	56	10.1	NA	13.1
3	2022-02-25_NEC	2022-02-25 00:00:00	NEC	ostr~	0	8.75	44.1	9.87
4	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.267	NA	NA	NA
5	2022-11-12_NEC	2022-11-12 00:00:00	NEC	ostr~	0	7.33	7.01	2.4
6	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
7	2021-11-16_NPB	2021-11-16 00:00:00	NPB	ostr~	72.1	7.3	2.38	12.1
8	2022-05-06_NMC	2022-05-06 00:00:00	NMC	ostr~	0	9.96	1.45	16.5
9	2023-02-21_NEC	2023-02-21 00:00:00	NEC	ostr~	0	8.56	NA	8.8
10	2023-09-12_NVBR	2023-09-12 00:00:00	NVBR	ostr~	0	9.14	NA	20.3

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
```



```
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Coding the figure

```
# base layer: x = site, y = log transformed ostracod abundance
ostracods_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +

# individual survey observations as open circles
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +

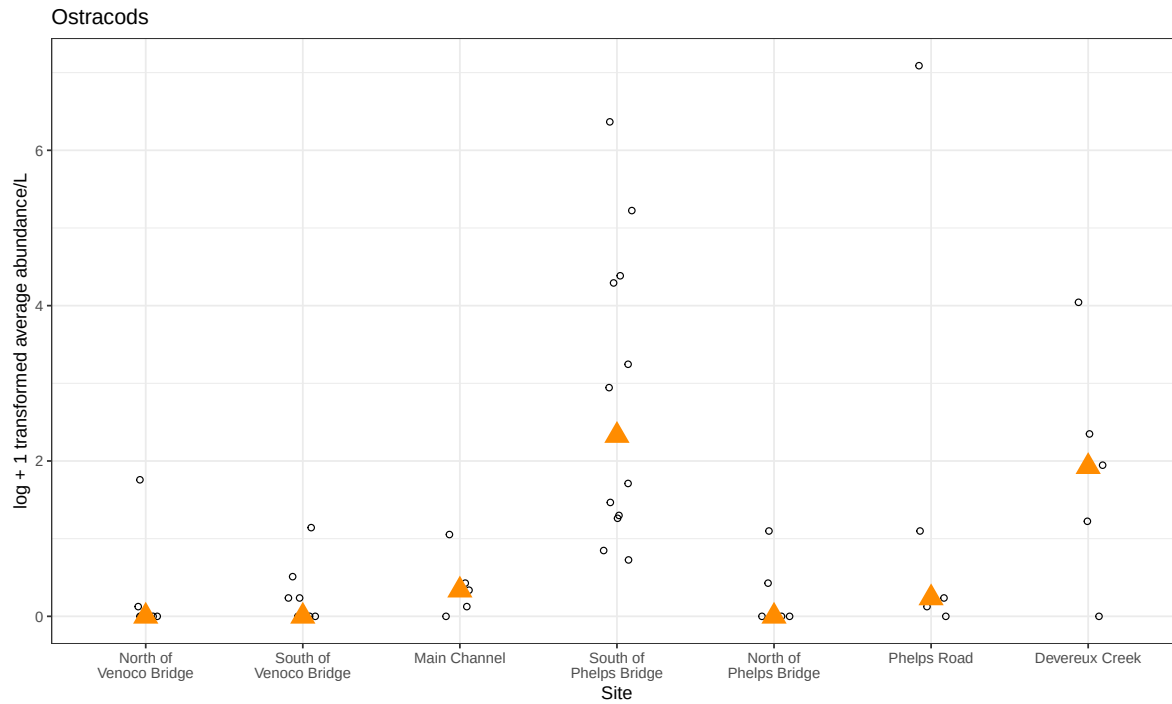
# median per site: large filled orange triangle to stand out from
# observations
stat_summary(geom = "point",
            fun = "median",
            size = 5,
            color = "darkorange",
            shape = 17) +

# wrap long site names so they don't overlap on x-axis
scale_x_discrete(labels = label_wrap(15)) +

# clear axis labels and title naming the taxonomic group
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +

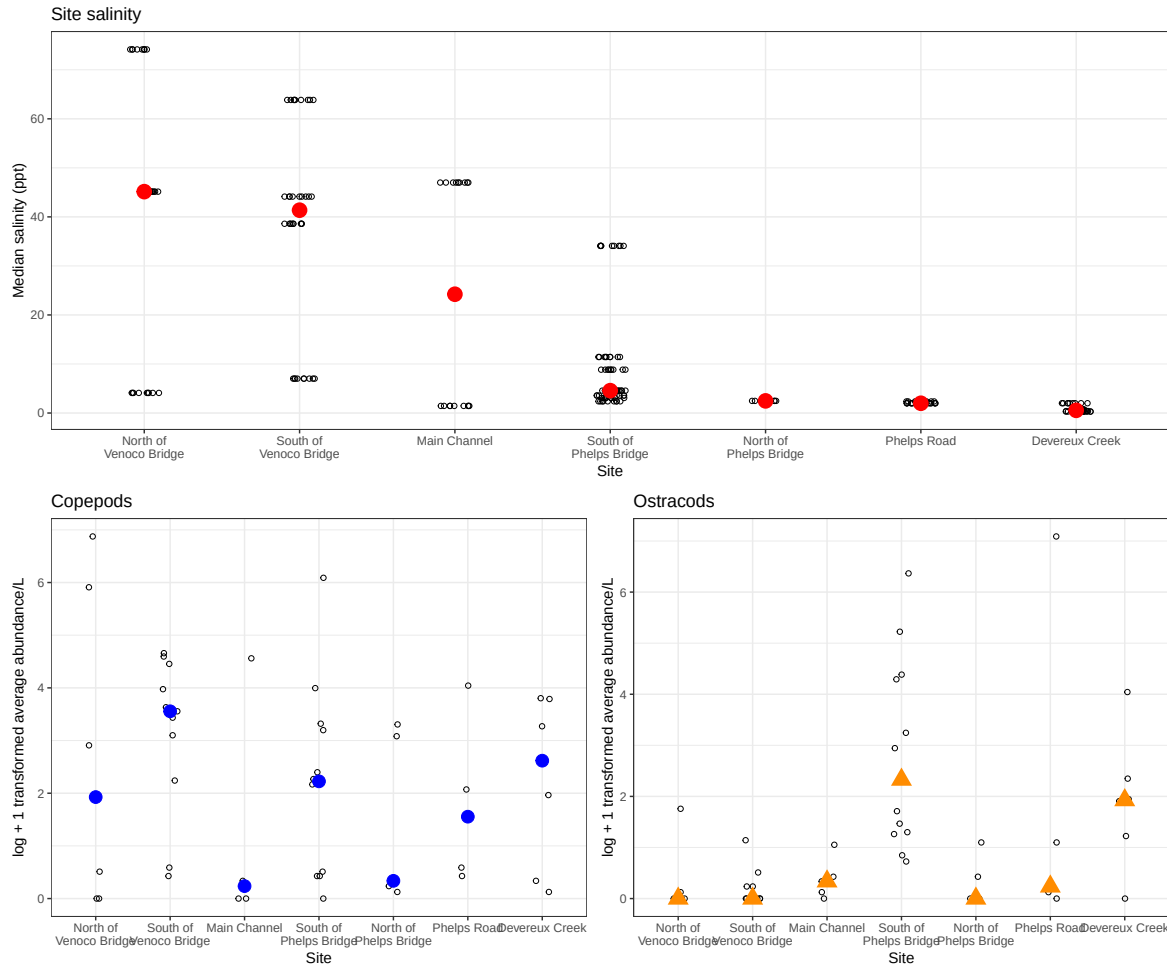
# clean theme different from default gray
theme_bw()

# display the plot
ostracods_plot
```



d. Create a multipanel figure

```
# salinity on top spanning full width, copepods and ostracods side by side
↔ below
salinity / (copepods_plot | ostracods_plot)
```



e. Interpret your figure

Copepod and Ostracod abundance differ across sites, copepod median abundance depicted by the blue points is highest at South Venoco bridge with a median of 3.5, and Devereux creek with a median of approximately 2.7, with it's lowest median being at Main channel and North of Phelps Bridge with a median of approximately 0.3. Ostracod median abundance represented by the orange triangles is highest at South of Phelps Bridge with a median of approximately 2.3, while most other sites have medians near zero.

The highest Ostracod abundance ~ 2.3 at south of Phelps Bridge has relatively low median salinity depicted by the red point approximately 5 ppt. North and South of Venoco Bridge have the highest salinity medians, 45, and 41 ppt respectively and have near zero medians which suggests that Ostracods may be negatively impacted by high salinity conditons.

2. Visualizing total abundance as a function of salinity

a. Planning my figure

-x-axis: `med_sal` (median salinity in ppt, continuous) - y-axis: `log_ave_lit` (log + 1 transformed average abundance per liter) - color: `taxon` (one color per taxon) - geometry: `geom_point()` — one point per survey event per taxon - facets: `facet_wrap(~ taxon, scales = "free")` — separate panel per taxon with free x and y axis scales

b. Wrangle your data

- log + 1 transform average abundance per liter
- convert taxon names to title case
- order taxon levels by maximum abundance

```
# create new data frame from aq_ins_clean
sal_abundance <- aq_ins_clean |>
  # log + 1 transform average abundance per liter to reduce skew
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon names to title case for readable panel labels
  mutate(taxon = str_to_title(str_replace_all(taxon, "_", " "))) |>
  # order taxon factor levels by maximum abundance (highest max first)
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = "max", .desc = TRUE))

# display 10 random rows to verify output
slice_sample(sal_abundance, n = 10)
```

A tibble: 10 x 24

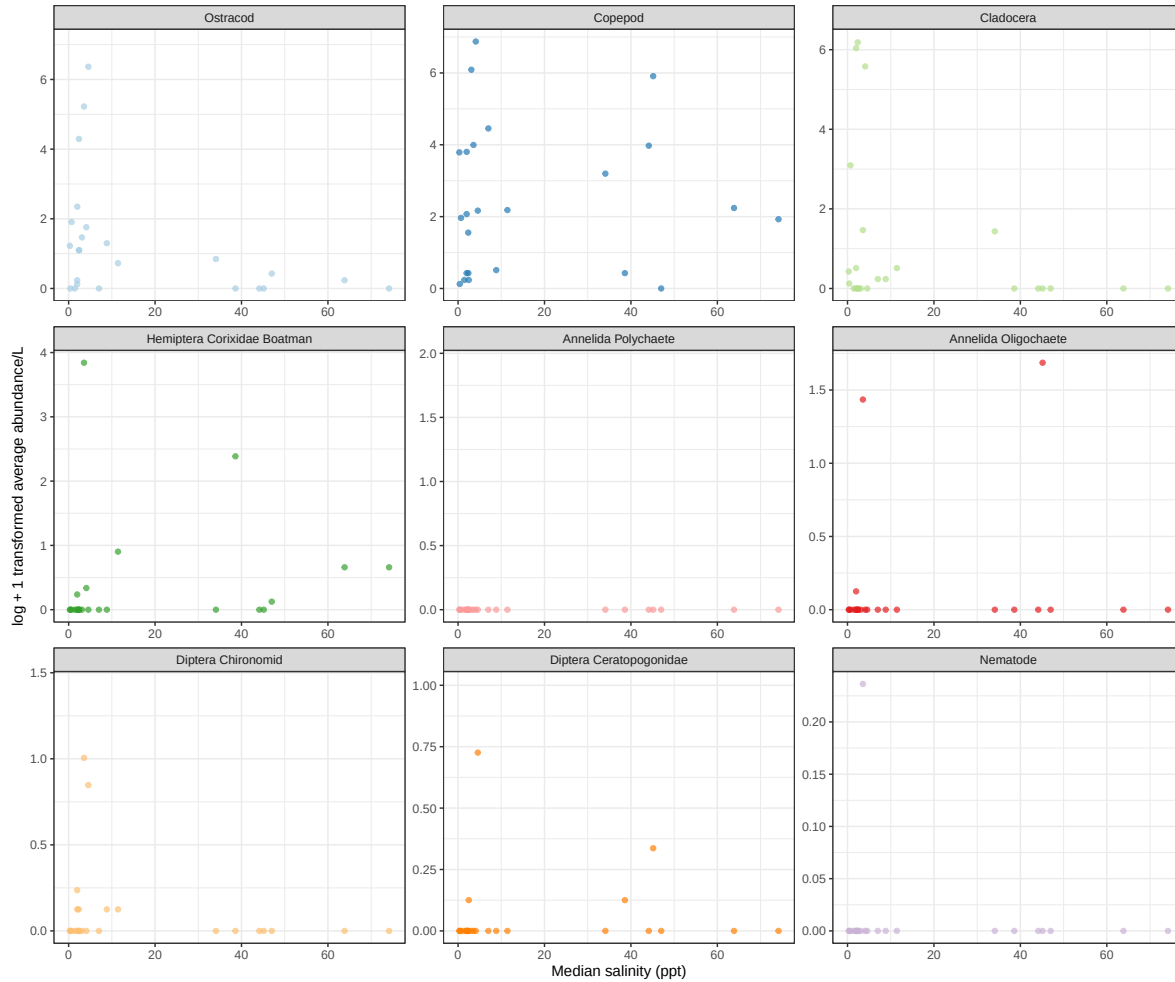
	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-05-04_NMC	2023-05-04 00:00:00	NMC	Anne~	0	8.32	NA	10.3
2	2023-02-21_NEC	2023-02-21 00:00:00	NEC	Clad~	0.4	8.56	NA	8.8
3	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	Cope~	5.87	9.72	74.2	11.4
4	2023-08-12_NEC	2023-08-12 00:00:00	NEC	Nema~	0	NA	NA	NA
5	2022-05-06_NMC	2022-05-06 00:00:00	NMC	Ostr~	0	9.96	1.45	16.5
6	2023-05-04_NEC	2023-05-04 00:00:00	NEC	Dipt~	0	8.05	NA	8.55
7	2024-01-28_NPB	2024-01-28 00:00:00	NPB	Anne~	0	7.43	3.08	6.6
8	2022-01-21_NEC	2022-01-21 00:00:00	NEC	Dipt~	0	NA	NA	NA
9	2023-02-28_NPB	2023-02-28 00:00:00	NPB	Cope~	8.67	NA	NA	NA
10	2023-03-03_NDC	2023-03-03 00:00:00	NDC	Clad~	21.1	7.45	0.688	5.64

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,

```
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# base layer: x = salinity, y = log transformed abundance, color = taxon
ggplot(data = sal_abundance,
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +
  # one point per survey event per taxon
  geom_point(alpha = 0.7) +
  # separate panel per taxon with free x and y scales
  facet_wrap(~ taxon, scales = "free") +
  # non-default color palette
  scale_color_brewer(palette = "Paired") +
  # clear axis labels
  labs(x = "Median salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +
  # clean theme different from default gray
  theme_bw() +
  # remove legend since taxon is already labeled in facet strips
  theme(legend.position = "none")
```



d. Interpreting the figure

Ostracod, copepod and cladocera show the highest log-transformed abundance values overall with most of their higher abundance observations at low salinity under 10 ppt. Copepods have some higher abundance points at mid range salinity levels approximately around 40 ppt. Annelida polychaete, diptera chironomid, diptera ceratopogonidae, and nematode have low abundance across all salinity levels, with many points clustered near zero regardless of salinity, suggesting that their presence is just rare at NCOS. Annelida oligochaete shows an interesting pattern peaking in abundance at higher salinity values between 45-70 ppt which could suggest that they are pretty tolerant to salty conditions.